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The case for broader use of HOMA-IR in pediatric metabolic assessment: a narrative review

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The reliance on Body Mass Index (BMI) and fasting glucose as primary indicators of pediatric metabolic health may systematically fail to detect the earliest phase of metabolic dysfunction: chronic hyperinsulinemia. This narrative review, conducted in accordance with the Scale for the Assessment of Narrative Review Articles (SANRA) framework, synthesises current literature to evaluate the evidence for the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) as a potential first-line tool in pediatric metabolic assessment. We examine the physiological mechanisms of early-onset insulin resistance, the epigenetic and intergenerational consequences of obesogenic environments, the limitations of current diagnostic criteria, and the economic implications of delayed detection. We discuss both universal and targeted screening strategies, comparing HOMA-IR with alternative insulin resistance indices including the Quantitative Insulin Sensitivity Check Index (QUICKI), the Triglyceride-Glucose (TyG) index, and the Single Point Insulin Sensitivity Estimator (SPISE). We critically appraise the principal limitations of HOMA-IR—including insulin assay variability, pubertal physiological variation, and the absence of prospective screening trial data demonstrating improved clinical outcomes. We conclude that while the mechanistic and epidemiological evidence supporting HOMA-IR is substantial, prospective population-level screening trials are required to confirm that HOMA-IR-guided intervention improves clinical outcomes at scale, and we identify such trials as a research priority.

KEYWORDS

childhood metabolic syndrome, HOMA-IR, insulin resistance indices, metabolic screening, pediatric insulin resistance, precision pediatric medicine, pubertal insulin resistance, targeted screening

1 Introduction

1.1 Review methodology

This article is a narrative review conducted in accordance with the SANRA (Scale for the Assessment of Narrative Review Articles) framework for structured narrative reviews. A systematic literature search was conducted in PubMed/MEDLINE, Scopus, and Google Scholar using the following search terms, applied individually and in combination: “HOMA-IR children,” “HOMA-IR pediatric,” “insulin resistance screening children,” “pediatric metabolic syndrome HOMA-IR,” “fasting insulin reference values children,” “childhood insulin resistance cutoff,” “HOMA-IR pubertal,” “pediatric type 2 diabetes screening,” and “HOMA-IR vs. HbA1c children.”

The search was not restricted by publication date but was weighted toward studies published from 2010 onwards to reflect the contemporary evidence base. Studies were

included if they: (1) reported HOMA-IR values or cutoffs in pediatric populations (age ≤ 18 years); (2) compared HOMA-IR with at least one alternative metabolic screening tool; (3) reported outcomes related to metabolic syndrome, insulin resistance, Non-Alcoholic Fatty Liver Disease (NAFLD), or type 2 diabetes (T2D) risk in children; or (4) provided normative reference data for HOMA-IR or related indices in pediatric populations. Studies were excluded if they were case reports, conference abstracts without full-text availability, or studies conducted exclusively in adult populations without pediatric subgroup data. In addition to the primary literature search, reference lists of identified systematic reviews and meta-analyses were hand-searched to identify additional relevant studies. Clinical practice guidelines from the American Diabetes Association (ADA), the International Society for Pediatric and Adolescent Diabetes (ISPAD), the American Academy of Pediatrics (AAP), and the National Institute for Health and Care Excellence (NICE) were reviewed to contextualise the current standard-of-care recommendations.

The narrative synthesis was structured around six thematic domains: (1) the pathophysiology of early-onset insulin resistance; (2) the evidence base for HOMA-IR as a screening tool; (3) the limitations and controversies surrounding HOMA-IR; (4) comparison with alternative insulin resistance indices; (5) the economic and public health implications of delayed detection; and (6) the therapeutic and implementation implications of early HOMA-IR identification. Where the evidence is primarily mechanistic or derived from modelling studies, this is explicitly noted and distinguished from evidence derived from prospective clinical trials.

1.2 The current screening landscape

The current pediatric screening model relies heavily on BMI and fasting glucose. However, these markers have well-documented limitations. BMI cannot distinguish between metabolically healthy lean mass and metabolically adverse visceral adiposity; a child with a “normal” BMI may harbour significant visceral fat, chronic low-grade inflammation, and

insulin resistance (2). Fasting glucose becomes abnormal only after prolonged compensatory hyperinsulinemia has exhausted pancreatic beta-cell reserve—it reflects late-stage glycaemic dysregulation rather than early metabolic risk (3).

1.2.1 Critical evidence gap

It must be stated at the outset that, as of the time of writing, no prospective population-level screening trial has demonstrated that routine HOMA-IR screening improves clinical outcomes including type 2 diabetes incidence, Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) progression, cardiovascular events, quality of life, or long-term mortality. The arguments presented in this review are therefore positioned as an evidence-based rationale for such trials, not as a claim that current evidence is sufficient to mandate universal adoption. This limitation directly affects the strength of the recommendations that follow and should be borne in mind throughout.

1.3 Universal versus targeted screening: a balanced perspective

A central question for clinical implementation is whether HOMA-IR should be incorporated into universal screening of all children or targeted screening of high-risk populations. Current pediatric practice generally focuses metabolic assessment on children with established risk factors (19), including:

- Obesity (BMI ≥ 95 th percentile) or severe overweight (BMI 85th–95th percentile)
- Acanthosis nigricans
- Family history of type 2 diabetes (first- or second-degree relative)
- Polycystic Ovary Syndrome (PCOS) or clinical hyperandrogenism
- MASLD/NAFLD
- Small for gestational age (SGA) birth history
- Chronic glucocorticoid exposure
- Selected genetic syndromes associated with insulin resistance (e.g., Turner syndrome, Prader-Willi syndrome)

A targeted screening strategy—incorporating HOMA-IR into the metabolic assessment of children with one or more of these risk factors—represents a pragmatic, lower-cost first step that could be implemented within existing clinical workflows without requiring new infrastructure. This approach would identify the majority of children with clinically significant insulin resistance while avoiding the challenges of universal screening, including the potential for over-diagnosis in low-risk populations and the downstream resource implications of false-positive results.

However, a targeted-only approach has limitations. The concept of the “metabolically obese, normal-weight” child—in whom visceral adiposity and insulin resistance exist despite a normal BMI (2)—suggests that risk-factor-based screening will inevitably miss some affected individuals. Whether the prevalence of insulin resistance in the general pediatric

Abbreviations

AAP, american academy of pediatrics; ADA, american diabetes association; ALT, alanine aminotransferase; BMI, body mass index; CGM, continuous glucose monitoring; EASD, european association for the study of diabetes; FG, fasting glucose; FI, fasting insulin; GH, growth hormone; GLP-1, glucagon-like peptide-1; GTT, glucose tolerance test; HbA1c, glycated haemoglobin; HDL, high-density lipoprotein cholesterol; HOMA-B, homeostatic model assessment for beta-cell function; HOMA-IR, homeostatic model assessment for insulin resistance; hs-CRP, high-sensitivity C-reactive protein; IFCC, international federation of clinical chemistry and laboratory medicine; IGF-1, insulin-like growth factor 1; IR, insulin resistance; ISPAD, international society for pediatric and adolescent diabetes; MASLD, metabolic dysfunction-associated steatotic liver disease; NAFLD, non-alcoholic fatty liver disease; NASH, non-alcoholic steatohepatitis; NICE, national institute for health and care excellence; OGTT, oral glucose tolerance test; PCOS, polycystic ovary syndrome; PES, pediatric endocrine society; QUICKI, quantitative insulin sensitivity check index; SANRA, scale for the assessment of narrative review articles; SGA, small for gestational age; SGLT-2, sodium-glucose co-transporter 2; SPISE, single point insulin sensitivity estimator; T2D, type 2 diabetes; TG, triglycerides; TNF- α , tumour necrosis factor alpha; TyG, triglyceride-glucose index; WG-SIA, working group on standardisation of insulin assays.

population is sufficient to justify universal screening remains an empirical question that prospective studies must address.

This review presents the evidence supporting HOMA-IR as a screening tool without advocating definitively for either universal or targeted implementation. The optimal strategy will depend on population-specific prevalence data, healthcare system resources, and—most critically—the results of prospective screening trials demonstrating clinical benefit.

2 The hyperinsulinemia cascade in the developing body

2.1 Pathophysiology of early-onset insulin resistance

The first detectable sign of metabolic dysfunction is not elevated blood sugar; it is elevated fasting insulin. In a child consuming a diet high in ultra-processed foods and refined carbohydrates, repeated postprandial glucose excursions stimulate sustained pancreatic insulin secretion (4). Over time, peripheral tissues (primarily skeletal muscle and liver) downregulate insulin receptor sensitivity, creating a state of insulin resistance (1). The pancreas compensates by increasing insulin output, establishing a cycle of chronic hyperinsulinemia.

This state of chronic hyperinsulinemia is particularly consequential in a developing body, where metabolic infrastructure is still maturing. The hyperinsulinemia cascade may drive several pathological processes simultaneously: hepatic *de novo* lipogenesis, potentially leading to NAFLD or MASLD, which may progress to steatohepatitis (NASH) (5); adipose tissue dysfunction, with expansion beyond physiological capacity and release of pro-inflammatory cytokines (TNF- α , IL-6) that may promote systemic inflammation (6); hormonal disruption, in which chronic insulin elevation may interfere with the hypothalamic-pituitary axis, potentially contributing to early-onset PCOS in girls and altered pubertal timing in boys (7); and neuroinflammation, in which systemic inflammation may cross the blood-brain barrier, potentially contributing to behavioural and cognitive effects (8).

Crucially, longitudinal studies demonstrate that insulin resistance begins to rise in mid-childhood, several years before the onset of puberty (9). This pre-pubertal rise highlights a potential window where intervention could prevent the compounding effects of pubertal insulin resistance.

2.2 The natural history of metabolic dysfunction

Figure 1 (proposed) illustrates the progressive natural history of metabolic dysfunction from initial hyperinsulinemia through to established type 2 diabetes:

Stage 1: Chronic Hyperinsulinemia (normal glucose, elevated fasting insulin) → **Stage 2:** Established Insulin Resistance (elevated HOMA-IR, compensated glycaemia) → **Stage 3:** Beta-cell Functional Decline (impaired fasting glucose, impaired glucose tolerance) → **Stage 4:** Prediabetes (HbA1c 5.7%–6.4%) → **Stage 5:** Type 2 Diabetes (HbA1c $\geq$ 6.5%)

HOMA-IR is positioned to detect dysfunction at Stages 1–2, whereas fasting glucose and HbA1c become abnormal only at Stages 3–5. This temporal advantage represents the principal clinical rationale for HOMA-IR screening.

3 Epigenetic programming and intergenerational risk

Emerging evidence demonstrates that early-life metabolic environments may alter DNA methylation and histone modification patterns in ways that are transmitted across generations (10–12). The primary mechanism involves insulin-driven alterations to the methylation of genes regulating adipogenesis, hepatic lipid metabolism, and pancreatic beta-cell development. Animal models have demonstrated that offspring of hyperinsulinemic mothers exhibit altered expression of key metabolic genes from birth, independent of postnatal dietary exposure (11). Human cohort data, while more limited, suggest that children born to mothers with gestational insulin resistance exhibit higher HOMA-IR values in early childhood than children born to metabolically healthy mothers, even after controlling for postnatal diet and BMI (12).

The clinical implication is that HOMA-IR screening in children may identify not only those at individual risk, but those whose risk is epigenetically amplified by maternal metabolic history—a population that may require earlier and more intensive monitoring. However, it should be noted that the epigenetic evidence is primarily derived from animal models and observational human cohorts; direct causal evidence in human populations remains limited.

4 HOMA-IR: evidence base and clinical application

4.1 The HOMA-IR formula

The Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) is calculated as follows:

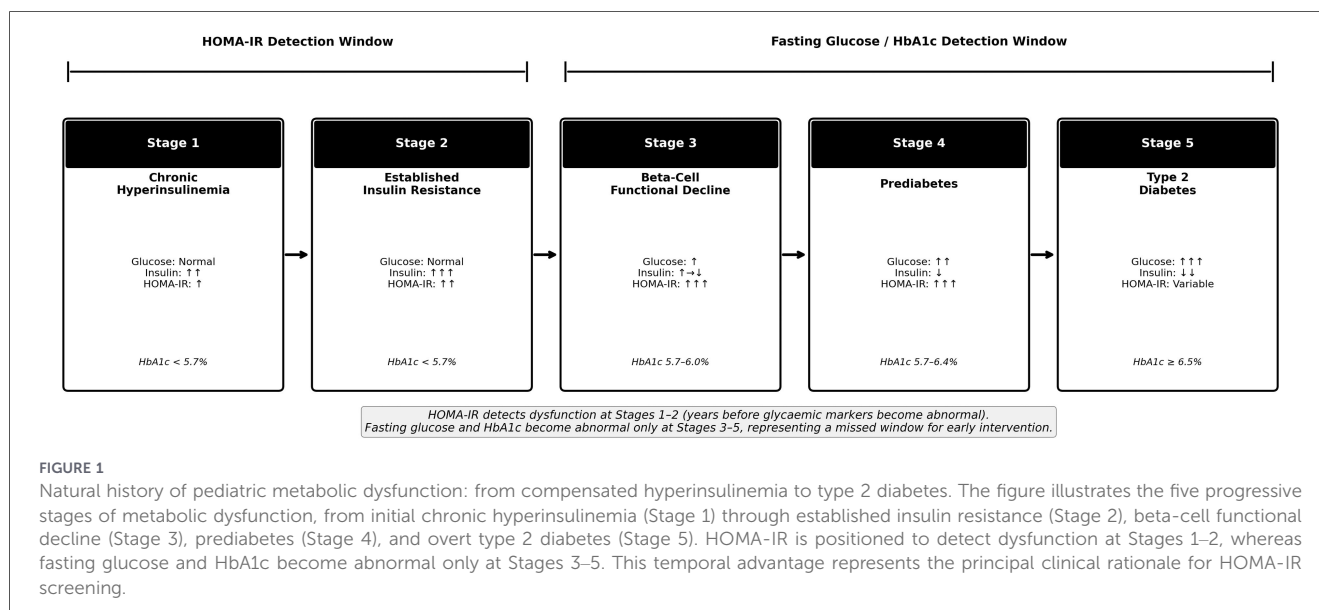
Using mg/dL units: $\text{HOMA-IR} = (\text{Fasting Insulin } [\mu\text{IU/mL}] \times \text{Fasting Glucose } [\text{mg/dL}]) / 405$.

Using mmol/L units: $\text{HOMA-IR} = [\text{Fasting Insulin } (\mu\text{IU/mL}) \times \text{Fasting Glucose } (\text{mmol/L})] / 22.5$.

HOMA-IR quantifies the degree of basal (primarily hepatic) insulin resistance from a single fasting blood sample (13). While the hyperinsulinemic-euglycaemic clamp remains the gold standard for measuring insulin sensitivity, it is highly invasive and impractical for clinical screening. HOMA-IR provides a validated, cost-effective surrogate that correlates well with clamp-derived measures in pediatric populations (13).

4.2 Establishing pediatric reference ranges

The scientific literature supports the utility of HOMA-IR in children, though establishing universal cutoffs has been challenging due to physiological changes during puberty. Consensus from multiple large-scale cohorts suggests that values



>2.5 consistently correlate with increased metabolic risk in prepubertal children (15, 16). However, these values must be interpreted with the following caveats:

Important interpretive considerations:

1. **Age dependence:** HOMA-IR values increase physiologically during puberty and decline in late adolescence.
2. **Pubertal stage:** Values during Tanner stages II–IV are physiologically higher than prepubertal or post-pubertal values.
3. **Sex differences:** Girls typically exhibit higher HOMA-IR values during puberty than boys of the same age.
4. **Ethnicity:** Population-specific reference ranges are required; cutoffs derived from European Caucasian populations may not apply to other ethnic groups (28).
5. **Insulin assay methodology:** Different immunoassays can yield insulin values differing by 20%–40% from the same sample (26).
6. **Laboratory calibration:** Results from different laboratories are not directly comparable.

The application of age-, sex-, and pubertal-stage-specific reference ranges is therefore essential for responsible clinical use. The large-scale study by Hammel et al. (31) provides age-stratified normative data from early childhood through adulthood (31), and recent data from Acta Paediatrica provide practical clinical thresholds for children with excess weight stratified by age, sex, and pubertal stage (32).

It must be explicitly acknowledged that, as of the time of writing, no major international pediatric or endocrine society—including the AAP, ISPAD, or the Pediatric Endocrine Society (PES)—has issued a formal recommendation for the routine clinical use of HOMA-IR as a standardised screening tool in children. This absence of formal endorsement reflects the evidence gap that this review seeks to highlight and that prospective screening trials must address.

4.3 Comparative screening performance

To contextualise the potential role of HOMA-IR within the broader screening landscape, Table 1 compares the principal metabolic screening tools currently available or under investigation in pediatric populations:

4.4 Comparison with alternative insulin resistance indices

Beyond the broader screening tools compared in Table 1, HOMA-IR must also be evaluated against other validated surrogate indices specifically designed to estimate insulin sensitivity (Table 2).

The TyG index has received particular attention as a potential alternative to HOMA-IR because it circumvents the insulin assay variability problem (29). Some studies suggest TyG may have comparable or even superior predictive value for metabolic syndrome in specific pediatric cohorts (30). However, TyG measures a downstream consequence of insulin resistance (hepatic lipid overflow manifesting as hypertriglyceridaemia) rather than the primary pathophysiological process (hyperinsulinemia). A child may exhibit clinically significant hyperinsulinemia (detectable by HOMA-IR) before triglyceride elevation occurs (detectable by TyG). This temporal difference suggests that HOMA-IR and TyG may be complementary rather than competing tools, with HOMA-IR detecting earlier-stage dysfunction.

The SPISE index, validated in European adolescent cohorts (34), offers the advantage of not requiring a fasting insulin measurement, but may be less sensitive to early-stage insulin resistance where lipid profiles remain normal.

4.5 HOMA-IR in the context of precision pediatric medicine

The emerging precision diabetology framework, as articulated by the ADA/EASD Consensus Report on Precision Medicine in

TABLE 1 Comparison of pediatric metabolic screening approaches.

Screening tool	Pathophysiological domain	Stage of dysfunction detected	Clinical feasibility	Cost	Pediatric evidence base
HOMA-IR	Basal hepatic insulin resistance	Stage 1—Hyperinsulinemia (pre-dysglycemia)	High—fasting blood draw only	Low (< \$20)	Strong—multiple large-scale cohorts (15–18)
Fasting Glucose	Beta-cell failure/glycaemic dysregulation	Stage 3—Established dysglycemia	High	Very low	Established but late-stage
HbA1c	Chronic glycaemic exposure	Stage 3—Established dysglycemia	High	Low	Established but late-stage
BMI	Total body mass (not metabolic)	Does not detect metabolic dysfunction	Very high	Negligible	Poor—misses metabolically obese normal-weight children
TyG Index	Hepatic insulin resistance (dyslipidemia)	Stage 2—Hepatic lipid overflow	High—fasting glucose + triglycerides	Low	Growing—several pediatric studies (29, 30)
Oral GTT	Dynamic glucose handling	Stage 2–3	Moderate—2-hour test required	Moderate	Strong but impractical for universal screening
CGM	Dynamic glucose variability	Stage 2–3	Low—device required	High	Emerging—not yet validated for universal pediatric screening

BMI, body mass index; CGM, continuous glucose monitoring; GTT, glucose tolerance test; HbA1c, glycated haemoglobin; HOMA-IR, Homeostatic Model Assessment for Insulin Resistance; TyG, Triglyceride-Glucose index.

TABLE 2 Comparison of surrogate insulin resistance indices applicable to pediatric populations.

Index	Formula/Derivation	What It measures	Advantages	Limitations	Pediatric evidence
HOMA-IR	$(FI \times FG)/405$	Basal hepatic IR	Simple, single fasting sample, extensive pediatric data	Assay-dependent, static measure	Strong—multiple large cohorts (15–18)
QUICKI	$1/(\log FI + \log FG)$	Basal insulin sensitivity	Mathematically equivalent to HOMA-IR; better distribution	Same limitations as HOMA-IR	Moderate—validated in children (33)
TyG Index	$\ln(TG \times FG/2)$	Hepatic lipid-driven IR	No insulin assay needed	Detects Stage 2 (lipid overflow), not Stage 1	Growing—several pediatric studies (29, 30)
SPISE	$600 \times HDL^{0.185} / (TG^{0.2} \times BMI^{1.338})$	Peripheral insulin sensitivity	No fasting insulin needed; validated in adolescents	Requires lipid panel; less sensitive to early IR	Moderate—validated in European cohorts (34)
Matsuda Index	Derived from OGTT	Whole-body insulin sensitivity	Captures dynamic glucose handling	Requires 2-hour OGTT; impractical for screening	Strong but impractical for universal use
Insulinogenic Index	$\Delta Insulin_{30} / \Delta Glucose_{30}$	Beta-cell secretory response	Assesses beta-cell function	Requires OGTT	Limited pediatric screening data
Disposition Index	Insulinogenic Index $\times$ Matsuda	Beta-cell compensation for IR	Integrates secretion and sensitivity	Requires OGTT	Research tool; not screening-applicable

BMI, body mass index; FG, fasting glucose; FI, fasting insulin; HDL, high-density lipoprotein cholesterol; HOMA-IR, Homeostatic Model Assessment for Insulin Resistance; IR, insulin resistance; OGTT, oral glucose tolerance test; QUICKI, Quantitative Insulin Sensitivity Check Index; SPISE, Single Point Insulin Sensitivity Estimator; TG, triglycerides; TyG, Triglyceride-Glucose index.

Diabetes, emphasises that metabolic disease encompasses biologically distinct subtypes requiring individualised approaches [ADA/EASD, 2020]. This raises an important question: is a single fasting-state surrogate sufficient to capture the full complexity of pediatric insulin resistance?

We acknowledge that HOMA-IR, as a basal hepatic insulin resistance surrogate, does not fully characterise the multidimensional nature of the insulin resistance syndrome, which also encompasses peripheral (skeletal muscle) insulin

resistance, adipose tissue dysfunction, and dynamic postprandial glucose handling. More sophisticated phenotyping tools—including oral glucose tolerance testing and emerging multi-omic approaches—may offer a more complete biological picture.

However, the practical reality of pediatric primary care demands a pragmatic first-line approach. HOMA-IR is positioned in this review not as a comprehensive phenotyping instrument, but as a potential clinical gateway to more detailed metabolic assessment—the tool that identifies which children

may require more sophisticated evaluation. This is analogous to the role of HbA1c in adult T2D screening: it is not a complete characterisation of the metabolic phenotype, but it may serve as an initial step that determines who enters the precision medicine pathway.

5 Limitations and controversies: a critical assessment

While the utility of HOMA-IR in pediatric metabolic assessment is increasingly recognised, its universal adoption has been hindered by several substantive limitations. Addressing these honestly is essential for responsible clinical implementation.

5.1 Insulin assay variability

The most significant barrier to establishing universal HOMA-IR cutoffs is the variability in fasting insulin measurement across different laboratories (20). Different immunoassays can yield insulin values that differ by 20%–40% from the same blood sample (26), meaning that a HOMA-IR value calculated from one laboratory's assay cannot be directly equated to a value from another.

5.1.1 Practical clinical implications

This variability means that:—Clinicians should utilise the same laboratory for serial measurements within an individual patient, prioritising the trajectory of HOMA-IR over any single absolute value.—Comparisons against published numerical thresholds (e.g., >2.5 or >3.5) must be interpreted with caution unless the threshold was derived using the same assay methodology.—Laboratory-specific reference ranges, ideally derived from local healthy pediatric populations, are more informative than universal numerical cutoffs.—Serial measurements performed within the same laboratory are considerably more clinically informative than single-point comparisons against universal thresholds.

The International Federation of Clinical Chemistry and Laboratory Medicine (IFCC), in collaboration with the ADA, maintains an active Working Group on Standardisation of Insulin Assays (WG-SIA) aiming to harmonise assays to a uniform reference system (27). Until universal standardisation is achieved, the clinical utility of HOMA-IR is best realised through within-patient longitudinal monitoring rather than rigid categorical classification.

5.2 Pubertal physiological insulin resistance

Puberty induces a state of transient, physiological insulin resistance that is essential to understand when interpreting HOMA-IR in pediatric populations. The following physiological considerations are critical:

5.2.1 Growth hormone-mediated insulin resistance

The primary driver of pubertal insulin resistance is the surge in growth hormone (GH) and insulin-like growth factor 1 (IGF-1) secretion during puberty. GH directly antagonises insulin signalling in peripheral tissues, reducing glucose uptake in skeletal muscle and increasing hepatic glucose output. This is a physiological adaptation that supports the anabolic demands of pubertal growth (14).

5.2.2 Sex differences

Girls typically develop higher HOMA-IR values during puberty than boys, reflecting greater physiological insulin resistance during female pubertal development. This sex difference peaks during mid-puberty (Tanner stages III–IV) and narrows in late adolescence (14).

5.2.3 Timing across tanner stages

Insulin sensitivity decreases by approximately 25%–50% during the pubertal transition, with the nadir occurring at Tanner stages II–III. The degree of physiological insulin resistance varies substantially between individuals (9, 14).

5.2.4 Interaction with obesity

In children with pre-existing obesity, pubertal insulin resistance compounds the pathological insulin resistance associated with adiposity, creating a “double hit” that may accelerate progression toward metabolic dysfunction. Distinguishing physiological from pathological insulin resistance in obese pubertal adolescents requires careful clinical judgement and longitudinal monitoring (15).

5.2.5 Spontaneous improvement after puberty

Physiological pubertal insulin resistance typically resolves spontaneously in late adolescence (Tanner stage V) as GH secretion normalises. HOMA-IR values that remain elevated beyond the pubertal transition are more likely to reflect pathological insulin resistance. This temporal pattern underscores the importance of longitudinal monitoring rather than single-point screening during puberty.

The solution is the application of age-, sex-, and pubertal-stage-specific reference ranges. The existence of physiological variation does not invalidate HOMA-IR as a screening tool; rather, it demands clinical nuance and the use of appropriate, stage-specific thresholds. Applying a static prepubertal cutoff (e.g., >2.5) to a pubertal adolescent will inevitably result in over-diagnosis.

5.3 Ethnic and racial variation

HOMA-IR cutoffs derived from European Caucasian populations may not accurately reflect metabolic risk in other

ethnicities. South Asian populations tend to develop insulin resistance and T2D at lower BMI and potentially lower HOMA-IR thresholds, while some African-American cohorts exhibit higher fasting insulin levels as a normal physiological variant (28). As with growth charts, the clinical application of HOMA-IR requires population-specific reference ranges. Large-scale cohorts are actively defining these percentiles globally, allowing for more precise, ethnically appropriate risk stratification.

5.4 Absence of prospective outcome data

The most fundamental limitation of the HOMA-IR screening argument is the absence of prospective evidence demonstrating that routine screening improves clinical outcomes. Specifically, no completed randomised controlled trial has demonstrated that HOMA-IR-guided screening and intervention reduces:

- Incidence of type 2 diabetes
- Progression of MASLD/NAFLD
- Cardiovascular events
- Quality of life
- Long-term mortality

This evidence gap is not unique to HOMA-IR—it applies to all proposed metabolic screening strategies in children—but it directly limits the strength of any recommendation for routine implementation. The arguments in this review are based on mechanistic plausibility, cross-sectional associations, and cost-effectiveness modelling, not on completed screening trials. Prospective validation remains the most urgent research priority.

6 The economic implications of delayed detection

6.1 Direct costs of pediatric metabolic disease

The lifetime direct medical costs of treating type 2 diabetes are estimated at over \$85,000 per individual, with the majority of costs stemming from the management of complications (21). When the disease begins in childhood, the duration of exposure to hyperglycaemia increases, leading to earlier and more severe microvascular and macrovascular complications, further inflating lifetime costs (22).

Conversely, the cost of adding a fasting insulin test to a routine pediatric blood panel is typically less than \$20 in most healthcare systems. A recent cost-effectiveness analysis suggested that utilising indices like HOMA-IR may allow for efficient resource allocation by identifying at-risk individuals before expensive pharmacological or surgical interventions are required (23).

6.2 Downstream costs of screening

However, a comprehensive cost-effectiveness assessment must also consider the downstream investigations and clinical activity generated by abnormal HOMA-IR results, including:

- Repeat laboratory assessments (confirmatory fasting insulin, glucose)
- Oral glucose tolerance testing for children with elevated HOMA-IR
- Hepatic imaging (ultrasound) for MASLD assessment
- Inflammatory biomarker panels (hs-CRP, ALT, lipid profile)
- Specialist referrals (paediatric endocrinology, hepatology)
- Follow-up visits and longitudinal monitoring

These downstream costs are difficult to estimate without prospective implementation data. A targeted screening strategy (Section 1.3) would substantially reduce these costs by limiting screening to children with established risk factors. The net cost-effectiveness of HOMA-IR screening—whether universal or targeted—remains an empirical question that implementation studies must address.

6.3 The window of reversibility

The principal economic argument for early detection is the concept of reversibility. Intensive lifestyle and dietary interventions—particularly the restriction of ultra-processed foods and free sugars—have been shown to reduce HOMA-IR and may reverse early-stage hepatic steatosis in children and adolescents (24, 25). Once pancreatic beta-cell function declines and T2D is established, achieving remission becomes substantially more difficult. Early identification through HOMA-IR may therefore open a therapeutic window in which low-cost lifestyle interventions can prevent the need for expensive long-term pharmacological management.

It is important to note that this cost-effectiveness argument is primarily derived from modelling studies and extrapolation from intervention trials; direct evidence from completed population-level HOMA-IR screening programmes remains limited.

7 From screening to action: therapeutic implications

7.1 Lifestyle and nutritional intervention as first-line response

The primary and most evidence-supported response to elevated HOMA-IR in children is targeted dietary and lifestyle modification. The restriction of ultra-processed foods, free sugars, and refined carbohydrates has been shown to reduce HOMA-IR and may reverse early-stage NAFLD within weeks of implementation (24, 25). Structured physical activity—particularly resistance training and high-intensity interval training—may improve peripheral insulin sensitivity through GLUT-4 upregulation in skeletal muscle, independent of weight loss [McCormack et al. (24)]. These interventions are non-pharmacological, low-cost, and reversible, and may be initiated upon HOMA-IR identification without specialist referral.

TABLE 3 Proposed expert-opinion therapeutic response framework for HOMA-IR-stratified pediatric patients.

HOMA-IR level	Proposed risk category	Proposed first-line response	Proposed second-line response	Specialist referral
<2.5 (prepubertal)	Low risk	Routine monitoring (annual)	—	Not indicated
2.5–3.5	Borderline elevated	Dietary counselling; reduce ultra-processed food intake; increase physical activity	Repeat HOMA-IR at 3 months	Consider if no improvement at 6 months
3.5–5.0	Elevated	Intensive dietary intervention; structured exercise programme; family-based lifestyle modification	Hepatic ultrasound; inflammatory biomarker panel	Paediatric endocrinology referral
>5.0	Severely elevated	Immediate specialist referral; consider metformin (age ≥ 10 with established T2D); oral GTT	Full metabolic panel; NAFLD assessment	Paediatric endocrinology + hepatology

GTT, glucose tolerance test; HOMA-IR, Homeostatic Model Assessment for Insulin Resistance; MASLD, Metabolic Dysfunction-Associated Steatotic Liver Disease; NAFLD, Non-Alcoholic Fatty Liver Disease; T2D, type 2 diabetes.

7.2 Pharmacological considerations

Metformin is the only pharmacological agent currently approved for use in children aged ≥ 10 years with established T2D in most jurisdictions. Its mechanism—primarily hepatic glucose production suppression and mild insulin sensitisation—is directly relevant to the HOMA-IR phenotype. However, the identification of elevated HOMA-IR before overt T2D onset may create a window in which lifestyle intervention precludes the need for pharmacological therapy entirely, which is the preferred clinical outcome.

SGLT-2 inhibitors, GLP-1 receptor agonists, and dual incretin therapies are not currently approved for pediatric use in most jurisdictions. However, their adult evidence base provides a rationale for future pediatric trials in HOMA-IR-stratified populations. A proposed expert-opinion therapeutic response framework for HOMA-IR-stratified pediatric patients is presented in Table 3.

7.3 Proposed therapeutic response framework

7.3.1 Critical interpretive note

The thresholds and management pathways in Table 3 represent the authors' proposed framework based on synthesis of the available literature. They are not validated clinical cut-offs and are not endorsed by any international pediatric or endocrine society. HOMA-IR values vary according to age, pubertal stage, sex, ethnicity, and insulin assay methodology. Clinicians should interpret these values in the context of age-, sex-, and pubertal-stage-specific reference ranges appropriate to their local population and laboratory. The management strategies proposed here should be considered as expert opinion requiring prospective validation, not as evidence-based clinical guidelines.

7.4 Proposed validation framework

The clinical utility of HOMA-IR-guided intervention could be validated through a prospective, randomised controlled trial in

which children identified as high-risk by HOMA-IR screening are randomised to standard care vs. HOMA-IR-guided intensive lifestyle intervention. Proposed primary endpoints would include HOMA-IR normalisation at 12 months; proposed secondary endpoints would include hepatic steatosis regression on ultrasound, hs-CRP reduction, and T2D incidence at 5-year follow-up. Such a trial would provide the direct clinical outcome evidence that currently represents the most significant gap in the HOMA-IR screening evidence base.

8 Clinical feasibility and implementation considerations

8.1 Laboratory infrastructure

HOMA-IR requires a fasting blood draw and a reliable insulin immunoassay—both of which are available in standard clinical laboratory settings in most high-income healthcare systems. The calculation itself is simple enough to be automated within any electronic health record system. We acknowledge that insulin assay availability and standardisation remain variable in low- and middle-income countries, and propose that investment in insulin assay infrastructure represents a priority in these settings.

8.2 Tiered implementation model

We propose a tiered implementation model for clinical use:

Tier 1 (Primary Care/Well-Child Visits): HOMA-IR as a first-line metabolic screen alongside standard blood glucose and lipid assessment. This tier requires no additional infrastructure beyond a fasting blood draw. Initially, this could be targeted to children with established risk factors (Section 1.3).

Tier 2 (Specialist Referral): For children with elevated HOMA-IR (>3.5 , adjusted for age and pubertal stage), additional assessments may include oral glucose tolerance testing, hepatic ultrasound, and inflammatory biomarker assessment (hs-CRP, ALT). DEXA scanning, salivary cortisol testing, and continuous glucose monitoring are not proposed as first-line screening tools; they are reserved for specialist-level assessment in children with confirmed metabolic risk.

9 Conclusion

This narrative review presents the evidence supporting broader use of HOMA-IR in pediatric metabolic assessment. The mechanistic rationale is substantial: HOMA-IR detects chronic hyperinsulinemia—the earliest measurable phase of metabolic dysfunction—years before fasting glucose or HbA1c become abnormal. The epidemiological evidence from multiple large-scale pediatric cohorts consistently demonstrates associations between elevated HOMA-IR and metabolic syndrome, NAFLD, and T2D risk. The tool is inexpensive, requires only a fasting blood sample, and can be integrated into existing clinical workflows.

However, the evidence base has important limitations that must be honestly acknowledged. No prospective population-level screening trial has demonstrated that HOMA-IR-guided screening and intervention improves clinical outcomes. The proposed thresholds are not validated clinical cut-offs. Insulin assay variability complicates cross-laboratory comparisons. Pubertal physiology requires careful interpretive nuance. The optimal screening strategy (universal vs. targeted) remains undetermined.

We therefore conclude that:

1. The mechanistic and epidemiological evidence supports HOMA-IR as a potentially valuable addition to pediatric metabolic assessment, particularly in children with established risk factors.
2. The integration of HOMA-IR into targeted screening of high-risk pediatric populations represents a pragmatic first step that could be implemented within existing clinical infrastructure.
3. The most urgent research priority is the design and funding of prospective, randomised screening trials—with HOMA-IR normalisation, NAFLD regression, and T2D incidence as primary endpoints—to determine whether HOMA-IR-guided intervention improves clinical outcomes at scale.
4. International standardisation of insulin assays remains essential for the responsible clinical application of HOMA-IR across healthcare systems.

The question is not whether HOMA-IR provides earlier detection of metabolic dysfunction than current screening tools—the evidence suggests it does. The question is whether earlier

detection translates into improved clinical outcomes, and this can only be answered by prospective trials.

Author contributions

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